

Description:

In the following set of problems, the notation R is used to refer to a *standard uniform* random variable (a continuous random value between 0 and 1) and r is its numerical value.

Part 1: Generate 10,000 random values r . For each r generated, calculate the random value X by: $X = -\ln(r)$, where “ln” is the natural logarithm function. Investigate the probability distribution of the random variable X by doing the following:

1. Create a relative frequency histogram of X .
2. Select a probability distribution that, in your judgement, is the best fit for X . Note that, in addition to selecting the distribution, you will also have to estimate the parameters of this distribution.
3. Support your assertion above by creating a probability plot for X .

Part 2: Generate three sets of standard uniform random values r_1 , r_2 , and r_3 , each consisting of 10,000 values. Next, calculate the random value X according to the following formula: $X = -\ln(r_1 r_2 r_3)$. Investigate the probability distribution of the random variable X by doing the following:

1. Create a relative frequency histogram of X .
2. Select a probability distribution that, in your judgement, is the best fit for X . Note that, in addition to selecting the distribution, you will also have to estimate the parameters of this distribution.
3. Support your assertion above by creating a probability plot for X .